

Maths — Phase automatismes — Corrigé

Calcul algébrique et trigonométrie

Exercice 1 — Sommes : linéarité et sommes usuelles

$$(a) \sum_{k=1}^n k(k+1) = \sum_{k=1}^n k^2 + \sum_{k=1}^n k = \frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2} = \frac{n(n+1)}{6} [(2n+1) + 3] = \boxed{\frac{n(n+1)(n+2)}{3}}.$$

(b) On développe :

$$\sum_{k=1}^n (2k-1)^2 = 4 \sum_{k=1}^n k^2 - 4 \sum_{k=1}^n k + \sum_{k=1}^n 1 = \frac{2n(n+1)(2n+1)}{3} - 2n(n+1) + n = \boxed{\frac{n(2n-1)(2n+1)}{3}}.$$

(Contrôle $n=2$: $1+9=10$ et $\frac{2 \cdot 3 \cdot 5}{3} = 10$.)

(c) On sépare la somme ($n+1$ termes) :

$$\sum_{k=0}^n 2^k = 2^{n+1} - 1, \quad \sum_{k=0}^n 4k = 2n(n+1), \quad \sum_{k=0}^n (n-3) = (n+1)(n-3),$$

$$\text{et } 2n(n+1) + (n+1)(n-3) = 3n^2 - 3, \text{ d'où } \sum_{k=0}^n (2^k + 4k + n - 3) = \boxed{2^{n+1} + 3n^2 - 4}.$$

(d) Somme géométrique de raison $2/3$:

$$\sum_{k=0}^n 2^k 3^{n-k} = 3^n \sum_{k=0}^n \left(\frac{2}{3}\right)^k = 3^n \cdot \frac{1 - (2/3)^{n+1}}{1 - 2/3} = \boxed{3^{n+1} - 2^{n+1}}.$$

Exercice 2 — Télescopage

$$(a) \frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}, \text{ donc } \sum_{k=1}^n \frac{1}{k(k+1)} = 1 - \frac{1}{n+1} = \boxed{\frac{n}{n+1}}.$$

$$(b) \sum_{k=1}^n \ln \frac{k+1}{k} = \ln \prod_{k=1}^n \frac{k+1}{k} = \boxed{\ln(n+1)} \text{ (produit télescopique } = n+1).$$

$$(c) 1 - \frac{1}{k^2} = \frac{(k-1)(k+1)}{k^2}, \text{ d'où}$$

$$\prod_{k=2}^n \left(1 - \frac{1}{k^2}\right) = \underbrace{\prod_{k=2}^n \frac{k-1}{k}}_{=1/n} \cdot \underbrace{\prod_{k=2}^n \frac{k+1}{k}}_{=(n+1)/2} = \boxed{\frac{n+1}{2n}}.$$

$$(d) \prod_{k=1}^n \frac{k+1}{k} = \boxed{n+1} \text{ (télescopique).}$$

Exercice 3 — Changement d'indice

(a) Posons $S = \sum_{k=1}^n k2^k$. Alors $2S = \sum_{k=1}^n k2^{k+1} = \sum_{j=2}^{n+1} (j-1)2^j$, donc

$$2S - S = \sum_{j=2}^{n+1} (j-1)2^j - \sum_{k=1}^n k2^k = n2^{n+1} - 1 \cdot 2^1 - \sum_{j=2}^n 2^j = n2^{n+1} - 2 - (2^{n+1} - 4),$$

soit $S = (n-1)2^{n+1} + 2$. $\left[\sum_{k=1}^n k2^k = (n-1)2^{n+1} + 2 \right]$ (contrôle $n = 2 : 2 + 8 = 10 = (1) \cdot 8 + 2$).

(b) Par le changement $j = n - k$, $\cos\left(\frac{(n-k)\pi}{2n}\right) = \cos\left(\frac{\pi}{2} - \frac{k\pi}{2n}\right) = \sin\left(\frac{k\pi}{2n}\right)$, donc

$$T_n := \sum_{k=0}^n \cos^2\left(\frac{k\pi}{2n}\right) = \sum_{k=0}^n \sin^2\left(\frac{k\pi}{2n}\right) \implies 2T_n = \sum_{k=0}^n 1 = n+1, \quad \left[T_n = \frac{n+1}{2} \right].$$

Exercice 4 — Sommes doubles

(a) À i fixé, j décrit $\{i, \dots, n\}$ ($n - i + 1$ valeurs) :

$$\sum_{1 \leq i \leq j \leq n} i = \sum_{i=1}^n i(n-i+1) = (n+1) \sum_{i=1}^n i - \sum_{i=1}^n i^2 = \left[\frac{n(n+1)(n+2)}{6} \right].$$

(b) $\sum_{1 \leq i, j \leq n} (i+j) = 2 \sum_{1 \leq i, j \leq n} i = 2n \sum_{i=1}^n i = 2n \cdot \frac{n(n+1)}{2} = \left[n^2(n+1) \right]$ (par symétrie $i \leftrightarrow j$).

(c) De $\left(\sum_{k=1}^n k\right)^2 = \sum_{i,j} ij = \sum_{i=j} i^2 + 2 \sum_{i < j} ij$, on tire

$$\sum_{1 \leq i < j \leq n} ij = \frac{1}{2} \left[\left(\frac{n(n+1)}{2} \right)^2 - \frac{n(n+1)(2n+1)}{6} \right] = \left[\frac{(n-1)n(n+1)(3n+2)}{24} \right].$$

(Contrôle $n = 3 : 1 \cdot 2 + 1 \cdot 3 + 2 \cdot 3 = 11$ et $\frac{2 \cdot 3 \cdot 4 \cdot 11}{24} = 11$.)

Exercice 5 — Coefficients binomiaux

On part de $(1+x)^n = \sum_{k=0}^n \binom{n}{k} x^k$.

(a) $x = 1 : \sum_{k=0}^n \binom{n}{k} = \left[2^n \right]$.

(b) $x = -1 : \sum_{k=0}^n (-1)^k \binom{n}{k} = (1-1)^n = \left[0 \right]$ pour $n \geq 1$ (et 1 si $n = 0$).

(c) La relation $k \binom{n}{k} = n \binom{n-1}{k-1}$ donne

$$\sum_{k=1}^n k \binom{n}{k} = n \sum_{k=1}^n \binom{n-1}{k-1} = n \sum_{j=0}^{n-1} \binom{n-1}{j} = \left[n 2^{n-1} \right].$$

(d) La relation $\frac{1}{k+1} \binom{n}{k} = \frac{1}{n+1} \binom{n+1}{k+1}$ donne

$$\sum_{k=0}^n \frac{1}{k+1} \binom{n}{k} = \frac{1}{n+1} \sum_{k=0}^n \binom{n+1}{k+1} = \frac{1}{n+1} \sum_{j=1}^{n+1} \binom{n+1}{j} = \boxed{\frac{2^{n+1} - 1}{n+1}}.$$

Exercice 6 — Linéarisation et expression

(a) $\cos^2 x \sin^2 x = \left(\frac{1}{2} \sin 2x\right)^2 = \frac{1}{4} \sin^2 2x = \frac{1}{4} \cdot \frac{1 - \cos 4x}{2} = \boxed{\frac{1}{8} - \frac{1}{8} \cos 4x}.$

(b) $\cos^3 x \sin x = \cos^2 x (\cos x \sin x) = \frac{1 + \cos 2x}{2} \cdot \frac{\sin 2x}{2} = \frac{1}{4} \sin 2x + \frac{1}{4} \sin 2x \cos 2x$, donc

$$\cos^3 x \sin x = \boxed{\frac{1}{4} \sin 2x + \frac{1}{8} \sin 4x}.$$

(c) Par la formule de Moivre, $\cos 5x = \Re((\cos x + i \sin x)^5) = \cos^5 x - 10 \cos^3 x \sin^2 x + 5 \cos x \sin^4 x$.
En posant $c = \cos x$ et $\sin^2 x = 1 - c^2$:

$$\cos 5x = c^5 - 10c^3(1 - c^2) + 5c(1 - c^2)^2 = \boxed{16 \cos^5 x - 20 \cos^3 x + 5 \cos x}.$$

(Contrôle $x = 0$: $16 - 20 + 5 = 1$.)

Exercice 7 — Sommes trigonométriques et équations

(a) Pour $x \not\equiv 0 [2\pi]$, en passant par $\sum_{k=0}^n e^{ikx} = \frac{e^{i(n+1)x} - 1}{e^{ix} - 1} = e^{inx/2} \frac{\sin \frac{(n+1)x}{2}}{\sin \frac{x}{2}}$ et en prenant la partie réelle :

$$\boxed{\sum_{k=0}^n \cos(kx) = \frac{\cos(\frac{nx}{2}) \sin(\frac{(n+1)x}{2})}{\sin(\frac{x}{2})}} \quad (x \not\equiv 0 [2\pi]),$$

et $\sum_{k=0}^n \cos(kx) = n+1$ si $x \equiv 0 [2\pi]$.

(b) $\cos(3x - \frac{\pi}{4}) = \cos \frac{\pi}{4} \iff 3x - \frac{\pi}{4} \equiv \pm \frac{\pi}{4} [2\pi]$:

$$3x = \frac{\pi}{2} + 2k\pi \text{ ou } 3x = 2k\pi \implies \boxed{x \in \left\{ \frac{\pi}{6} + \frac{2k\pi}{3} \right\} \cup \left\{ \frac{2k\pi}{3} \right\}, k \in \mathbb{Z}}.$$

(c) $\sin a = \sin b \iff a \equiv b [2\pi] \text{ ou } a \equiv \pi - b [2\pi]$, avec $a = x + \frac{\pi}{3}$, $b = 3x - \frac{\pi}{6}$:

(i) $x + \frac{\pi}{3} \equiv 3x - \frac{\pi}{6} \Rightarrow x \equiv \frac{\pi}{4} [\pi]$; (ii) $x + \frac{\pi}{3} \equiv \pi - 3x + \frac{\pi}{6} \Rightarrow x \equiv \frac{5\pi}{24} [\frac{\pi}{2}]$.

$$\boxed{x \in \left\{ \frac{\pi}{4} + k\pi \right\} \cup \left\{ \frac{5\pi}{24} + \frac{k\pi}{2} \right\}, k \in \mathbb{Z}}.$$